GeoEnergy Space Nexus Research Institute

Mission, Vision, and Core Values

Organizational Foundations Document
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This document defines the mission, vision, and core values that guide the work of the GeoEnergy Space Nexus Research Institute. It is formatted for inclusion in the organization's GitHub repository as part of its public documentation, in compliance with GitHub's community and content guidelines.

GeoEnergy Space Nexus
Research Institute

Our Mission

The mission of the GeoEnergy Space Nexus Research Institute is to advance open, collaborative research at the intersection of space science, geospatial technology, climate intelligence, and sustainable energy, while ensuring that the resulting knowledge and tools directly support humanitarian needs and resilient communities worldwide.

We pursue this mission by building accessible data infrastructure, developing open-source tools, and fostering partnerships across research institutions, humanitarian organizations, and local communities, with the goal of turning space-derived and spatial data into actionable knowledge for sustainable development.

Our Vision

We envision a world where space-based and spatial data are openly accessible and actively used to address humanity's most pressing challenges, climate adaptation, energy access, and humanitarian crises, through transparent, reproducible, and community-driven research.

The GeoEnergy Space Nexus Research Institute aspires to be a recognized hub for open geospatial and space research, known for the quality of its data, the impact of its tools, and the breadth of its collaborative network spanning research institutions, space agencies, humanitarian bodies, and local communities.

In the long term, we aim to demonstrate how open data and open-source collaboration, anchored in space science and geospatial intelligence, can accelerate progress toward sustainable energy systems, climate resilience, and humanitarian preparedness on a global scale.

GeoEnergy Space Nexus
Research Institute

Core Values

Openness

We are committed to open data, open-source software, and transparent research practices. We believe that knowledge generated through our work should be accessible to researchers, practitioners, and communities everywhere.

Collaboration

We value partnerships across disciplines, sectors, and borders. Our work is strengthened by contributions from research institutions, humanitarian organizations, government agencies, and the open-source community.

Integrity

We uphold rigorous scientific standards, ethical data practices, and honest reporting of methods and findings. Data provenance, privacy, and proper attribution are core to everything we publish.

Sustainability

We design our projects with long-term environmental and social impact in mind, prioritizing sustainable energy solutions, climate resilience, and responsible resource use.

Humanitarian Focus

We orient our research toward real-world impact, ensuring that data, tools, and findings can be applied to support vulnerable communities and humanitarian response efforts.

Innovation

We embrace exploration and experimentation, whether in space science, climate modeling, or energy planning, and we encourage creative, evidence-based approaches to complex problems.

Accessibility

We strive to make our datasets, tools, and documentation usable by researchers and practitioners at all levels, with clear documentation and inclusive design.

Strategic Goals

The following goals guide the implementation of our mission and vision across all project pillars and repositories.

Goal 1: Build Open Data Infrastructure

Establish and maintain well-documented, version-controlled repositories of space, spatial, humanitarian, and climate data, hosted and managed through GitHub in line with open data best practices.

Goal 2: Develop Open-Source Tools

Create and maintain open-source software for data processing, climate analytics, energy planning, and space exploration, with clear licensing and contribution pathways.

Goal 3: Strengthen Humanitarian Impact

Ensure that research outputs are connected to humanitarian use cases, supporting evidence-based decision-making in disaster response, resource allocation, and community resilience.

Goal 4: Advance Sustainable Energy Solutions

Provide geospatial tools and datasets that support the planning and expansion of renewable energy infrastructure, particularly in underserved regions.

Goal 5: Foster Open Collaboration

Grow an active community of contributors, researchers, and partner organizations engaged with the Institute's repositories, following clear contribution guidelines and codes of conduct.

Goal 6: Maintain GitHub Policy Compliance

Ensure that all repositories, documents, and published materials comply with GitHub's community guidelines, licensing requirements, and acceptable use policies.